

Nicholas (Nic) McDonald

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Expertise:

I am a computer architecture researcher and software/hardware engineer specializing in AI systems, accelerator design, and high-performance interconnects. As a Senior Research Scientist at NVIDIA, I develop analytical models and simulation tools to evaluate LLM training and inference across compute, memory, and communication. My work includes hardware/software co-design, energy-efficient inference architectures, and electrical and optical network design. Previously, at Google and Hewlett Packard Labs, I developed network architectures, routing algorithms, transport protocols, and congestion-control systems. I build tools that connect workload behavior to architectural decisions, including Calculon for LLM systems and SuperSim for interconnection networks.

Education:

Stanford University

- *Ph.D. Electrical Engineering, 2016, GPA 3.90*
 - *Advisor: William J. Dally*

University of Utah

- *M.S. Electrical and Computer Engineering, May 2012, GPA 3.85*
 - *Advisor: Al Davis*
- *B.S. Computer Engineering, December 2010, GPA 3.81*

Employment:

NVIDIA - Senior Research Scientist2021-present

- Developed Calculon, a framework for rapid design-space exploration of software/hardware co-design for large language models, enabling fast estimation of performance and resource usage across large execution spaces; published at SC23.
- Co-authored an analytical study of infrastructure for training LLMs with up to 128 trillion parameters, examining memory capacity, offloading, and system-scaling trade-offs; published at ASSYST 2023.
- Identified higher-efficiency LLM optimization and system design strategies through analytical modeling of software and hardware tradeoffs.
- Evaluated scale-up and scale-out interconnect and topology options for LLM workloads, including optical-circuit-switch-based approaches for efficient large-scale training.
- Contributed to distributed GEMM research and software infrastructure, including defining reusable multi-GPU APIs and helping integrate distributed algorithms into CUTLASS.
- Developed advanced AI accelerator design concepts focused on minimizing energy use and data movement while maximizing inference speed across emerging model workloads.

Google - Senior Software Engineer2019-2021

- Developed next generation data center network topologies, routing algorithms, transport protocols, congestion control systems, and switching architectures.
- Guiding the architecture of future accelerated network hardware and software systems.
- Developed ParaGraph, an application-simulator interface and toolkit for hardware/software co-design that connected compiler-derived workload models with simulation infrastructure for large-scale system exploration; published at ICPP 2022.

Hewlett Packard Labs - Research Scientist 2016-2018

- Developed architectures for next-generation HPC systems based on emerging technologies such as photonics, non-volatile memories, memory-semantic network protocols, etc.
- Designed the routing and congestion management architecture of the Gen-Z protocol to support HPC systems with efficiency at extreme scales and high performance.
- Designed a hierarchical high-radix router architecture for use in next-generation HPC systems.
- Designed new routing algorithms to support fine-grained incremental adaptive routing.

Google - Network Research Intern 2014

- Designed, prototyped, and benchmarked next generation network topologies, congestion control algorithms, and load balancing mechanisms.
- Developed a Python-based software package for interacting in parallel with numerous hosts with focus on error tolerability and scripting productivity.

HP Labs - Research Associate 2013-2014, 2015

- Designed multiple microarchitectures for high-radix switches enabled by photonic I/Os with emphasis on scalability of both monolithic and hierarchical switch designs.
- Developed a cycle-accurate discrete event network simulator specializing in large network simulations, flexible endpoint modeling, and alternative microarchitecture modeling.

L-3 Communications - Digital Hardware Design Engineer 2010-2012

- Developed digital processing architectures for encryption, networking, waveform data processing, and network synchronization.
- Designed systems with strict requirements for high-performance, low-power, and minimal area.
- All designs were meticulously scrutinized for high-reliability and signal integrity.

L-3 Communications - Systems Engineer 2008-2010

- Worked in systems engineering to support the Global Hawk UAV program.
- Designed digital test circuitry to aid product interfacing and testing.
- Developed acceptance test procedures for L-3 products.
- Performed many debugging investigations and tests for Global Hawk systems.

Publications:

“Calculon: a Methodology and Tool for High-Level Codesign of Systems and Large Language Models”, International Conference for High Performance Computing, Networking, Storage, and Analysis (SC), 2023.

“Scaling Infrastructure to Support Multi-Trillion Parameter LLM Training”, Workshop on Architecture and System Support for Transformer Models (ASSYST), 2023.

“ParaGraph: An application-simulator interface and toolkit for hardware-software co-design”, International Conference on Parallel Processing (ICPP), 2022.

“Practical and Efficient Incremental Adaptive Routing for HyperX Networks”, International Conference for High Performance Computing, Networking, Storage, and Analysis (SC), 2019.

“HyperX Topology: First At-Scale Implementation and Comparison to the Fat-Tree”, International Conference for High Performance Computing, Networking, Storage, and Analysis (SC), 2019.

“SuperSim: Extensible Flit-Level Simulation of Large-Scale Interconnection Networks”, International Symposium on Performance Analysis of Systems and Software (ISPASS), 2018.

“High-Performance Service-Oriented Computing”, Ph.D. dissertation, Stanford University, 2016.

Patents:

Named inventor on 20 granted U.S. patents covering network architecture, adaptive routing, congestion control, arbitration, and transport protocols.

"Rate update engine for reliable transport protocol", US11979330B2 (2024).

"Hierarchical switching device with deadlockable storage and storage partitions", US11855913B2 (2023).

"Route selection based on buffer congestion", US11722406B2 (2023).

"Predictive congestion detection", US11601368B2 (2023).

"Hot-swappable no cable touch switch enclosure", US11558682B2 (2023).

"Utilization of component group tables in a computing network", US11197152B2 (2021).

"Weighted bandwidth allocation for adaptive routing", US11178046B2 (2021).

"Network congestion management", US10972394B2 (2021).

"Hierarchical switching devices", US10938751B2 (2021).

"Prioritized arbitration using fixed priority arbiter", US10929322B2 (2021).

"Routing packets based on congestion metric thresholds and weights", US10887217B2 (2021).

"Reservation-based switching devices", US10757038B2 (2020).

"Request arbitration by age and traffic classes", US10693808B2 (2020).

"Routing packets using distance classes", US10498631B2 (2019).

"Virtual channel routing", US10491545B2 (2019).

"Routing packets based on congestion of minimal and non-minimal routes", US10476780B2 (2019).

"Switching device having ports that utilize independently sized buffering queues", US10404575B2 (2019).

"Routing packets in dimensional order in multidimensional networks", US10374943B2 (2019).

"Calculating times to live for transaction requests", US10355978B2 (2019).

"Arbitrating data packets", US9847949B1 (2017).

Open-Source Projects:

Calculon - LLM training/inference modeling and co-design, www.github.com/calculon-ai/calculon
Python, 4k lines, 25 files

ParaGraph - Application-simulator interface and toolkit, www.github.com/paragraph-sim
C++/Python, 36k lines, 145 files

SuperSim - Extensible flit-level network simulation, www.github.com/nicmcd/supersim
C++, 40k lines, 350 files

libdes - Parallel discrete event simulation framework, www.github.com/nicmcd/libdes
C++, 5k lines, 42 files

TaskRun - Parallel task management and execution, www.github.com/nicmcd/taskrun
Python, 6k lines, 44 files

ParaMgmt - Scriptable parallel server interaction, www.github.com/google/paramgmt
Python, 1k lines, 3 files

Academic Research:

A Distributed System Architecture for Secure High Performance Computing (Ph.D. thesis)

- *Developed a high performance distributed system architecture for modern large-scale applications that provides inherent network level isolation and security.*
- *Designed a network interface controller that supports this architecture by enforcing its policies efficiently in hardware and removing the need for complex software based solutions.*
- *Developed a custom logic simulator for the network interface controller and used it to show that it only increases message latency by 35-65 ns.*

Accelerator Multi-Threading (Multi-Core Architecture Class Project)

- *Developed a novel accelerator interface architecture for fine-grained system multi-threading.*
- *Implemented the design on a Xilinx Zynq platform and showed the dramatic benefits of hardware-based accelerator multi-threading versus software-based solutions.*

Arbitration Structures for High-Radix Routers (M.S. Research Project)

- *Implementation and investigation of the energy and delay scalability of parallel prefix arbitration structures for high-radix routers.*
- *Developed radix independent HDL and tested across multiple radices and technology libraries.*

Attack on SSL and the Public Key Infrastructure (Network Security Class Project)

- *Developed and implemented a system for covertly installing an evil certificate-authority (C.A.) in a victim's web browser.*
- *Developed software to reroute web traffic and use the evil C.A. to decrypt all transferred web data.*

Precise Time Synchronization in 802.15.4 Wireless Networks (Communications Class Project)

- *Developed digital hardware to precisely control time synchronization between nodes in an 802.15.4 wireless network.*

Fly-by-wire System for an Autonomous Unmanned Helicopter (B.S. Senior Project)

- *Developed a multi-processor system for sensor data acquisition, sensor fusion, and autonomous flight control using PID feedback.*
- *Created software for viewing and controlling the flight.*

Tiny 32-bit RISC Architecture (Processor Design Class Project)

- *Designed a programmer/compiler friendly instruction set architecture (I.S.A.).*
- *Created a fully featured assembler for machine code generation.*
- *Implemented and tested the processor design in an FPGA.*
- *Processor design includes a priority encoded interrupt system, variable sized instructions, SDRAM interface, and simple programming interface for precise hardware control.*